

TSS-LCD: A Temporal–Spectral–Spatial-Guided Latent Conditional Diffusion Model for Spectrum Prediction Under Incomplete Observations

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Abstract—Accurate spectrum prediction provides the foresight needed for timely access decisions and proactive interference avoidance in diverse wireless scenarios. Since spectrum RSS exhibits inherent, coupled dependencies across the TSS dimensions reflecting underlying spectrum usage patterns, most existing methods extract TSS features from complete historical observations and map them to future RSS through simple regression structures. However, in practical deployments the historical observations are often incomplete, which corrupts or removes informative patterns and makes it difficult to joint capture TSS dependencies. Furthermore, spectrum RSS data contains fine-grained variations; direct feature-to-prediction tends to smooth these details, reducing prediction accuracy. To address these problems, we propose TSS-LCD, a two-stage network that jointly captures TSS dependencies and then uses them to guide diffusion process in latent space for spectrum prediction. Specifically, the TSS-CC stage employs three parallel self-attention branches and a cross-attention fusion module to extract and integrate TSS dependencies into a unified conditional representation. The LCD-SP stage then performs latent conditional diffusion, using this representation as conditioning at every denoising step to reconstruct detail-preserving future RSS data. Experiments on a real-world dataset show that TSS-LCD outperforms representative baselines, achieving lower errors and better recovery of fine-grained variations under incomplete historical observations.

Index Terms—Spectrum prediction, deep learning, temporal-spectral-spatial (TSS) attention, latent diffusion model, incomplete observations.

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I. INTRODUCTION

WITH the explosive growth of wireless communication and the surge of spectrum-dependent applications, efficient and intelligent spectrum management has become essential across diverse wireless scenarios, such as public networks, industrial IoT, safety-critical services, and contested electromagnetic environments [2], [3], [4], [5], [6], [7], [8]. For example, next-generation systems (e.g., 6G, IoT, V2X) must make timely, adaptive access decisions under limited spectral resource; yet sense-before-access policies provide only instantaneous spectrum Received Signal Strength (RSS) data and cannot anticipate upcoming usage patterns, often leading to suboptimal channel selection and delayed interference avoidance [9], [10], [11], [12], [13]. Likewise, in interference-prone or adversarial conditions, many systems still react passively after anomalies become evident, risking severe performance loss or service interruption [14], [15]. Against this backdrop, *spectrum prediction*, i.e., forecasting future RSS from historical observations, supplies the missing foresight. It can serve as a building block for downstream spectrum management, and can support early warning by comparing observed RSS with predictions to identify abnormal behaviors such as unauthorized transmissions or jamming, thereby contributing to efficient spectrum utilization and resilient operation [16], [17], [18], [19].

Recently, numerous efforts have been devoted to the field of spectrum prediction. Traditional model-based methods, including autoregressive integrated moving average (ARIMA) models, hidden Markov models (HMMs), compressive sensing, and tensor completion methods, have demonstrated promising results under specific assumptions and constraints [20], [21], [22], [23], [24], [25]. These approaches generally assume that spectrum RSS data exhibits stationarity, periodicity, and low-rank properties, thereby inferring future spectrum RSS data based on historical observations using pre-defined statistical models. However, in wireless networks, spectrum RSS data are affected by a variety of factors, such as signal activity and environmental variations, which are highly dynamic and statistically non-stationary. More importantly, these factors generally exhibit complex and inherent correlations across the temporal, spectral, and spatial (TSS) dimensions, leading to multi-dimensional and coupled TSS dependencies within spectrum RSS data. Hence, effectively modeling such dependencies is crucial for capturing the underlying spectrum usage patterns, which further facilitates accurate spectrum prediction. However, the fixed-form, sta-

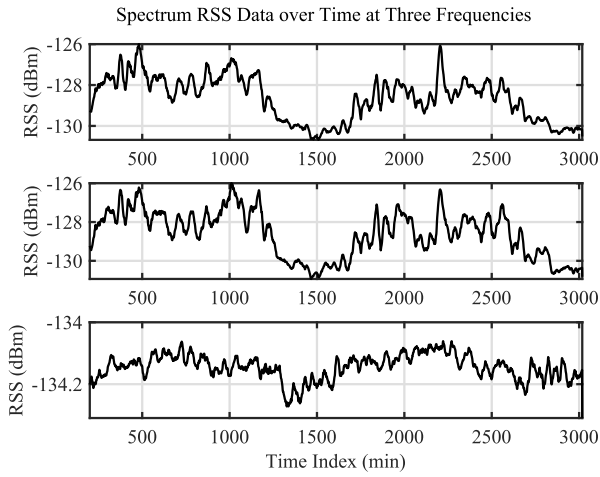


Fig. 1. Spectrum RSS data over time at three frequencies. Each trace illustrates rapid but low-amplitude fluctuations (typically within 2–5 dBm), highlighting the fine-grained structure in temporal dynamics.

tionary assumptions of these model-based methods render them unable to capture the non-stationary, multi-dimensional dependencies inherent in spectrum RSS data, resulting in limited prediction accuracy.

To address this limitation, deep learning (DL) techniques have emerged as powerful alternatives, offering strong capabilities in modeling non-linear and multi-dimensional dependencies directly from historical spectrum observations. Unlike traditional methods that rely on predefined priors, DL methods can automatically learn multi-dimensional dependencies by extracting informative features from historical observations, thereby enabling more expressive and accurate prediction. Motivated by this, a growing number of studies have explored the application of DL models for spectrum prediction. Several existing works focus exclusively on capturing temporal dependencies within spectrum RSS data, leveraging sequential models, such as recurrent neural networks (RNNs), long short-term memory networks (LSTMs), and Transformers, to extract multi-scale temporal features [26], [27], [28], [29], [30]. Some other studies further incorporate frequency or spatial dependencies into the modeling process, employing convolutional neural networks (CNNs) or graph convolutional networks (GCNs) to jointly extract temporal-frequency or temporal-spatial features [31], [32], [33], [34], [35], [36], [37], [38], [39], [40]. However, most of these approaches consider either temporal-frequency or temporal-spatial dependencies, without fully modeling the joint TSS dependencies within the historical spectrum RSS data.

Recent works have attempted to explicitly capture TSS dependencies in a unified manner, achieving improved prediction performance by jointly extracting multi-dimensional TSS features from complete historical observations [41], [42], [43]. While these methods demonstrate the potential of TSS dependencies modeling, they generally assume complete spectrum observations across all TSS dimensions. However, due to hardware constraints, energy limitations, or sensing strategies (e.g., partial-band sensing), spectrum observations are sparse and irregular. Under such incomplete observations, missing

entries can break TSS correlations, making it difficult to extract reliable TSS dependencies because informative usage patterns may be distorted or entirely absent. Furthermore, most existing methods [41], [42], [43] rely on classic regression task architectures, such as fully connected (FC) layers, to directly map the extracted features to future predictions. Such architectures typically lack sufficient capacity to retain fine-grained spectrum details. However, spectrum RSS data generally exhibit rapid yet low-amplitude fluctuations (typically vary within a narrow range of 2 to 5 dBm as shown in Fig. 1), indicating subtle but critical variations across TSS dimensions. Therefore, accurately reconstructing these fine-grained variations is essential for reliable spectrum prediction, yet inherently challenging, as they are easily smoothed out or suppressed by models lacking sufficient detail-preserving capacity.

Considering the aforementioned issues, in this paper, we propose a Temporal-Spectral-Spatial Guided Latent Conditional Diffusion Model (TSS-LCD), a two-stage network designed to accurately predict spectrum RSS data that exhibits fine-grained variations under incomplete historical observations. In our previous work [1], we proposed a multi-scale temporal model for 3D spatial spectrum prediction, focusing on capturing multi-scale temporal dependencies under complete observations. Different from our previous work, this work explicitly considers incomplete historical measurements and aims to obtain accuracy and reliable predictions from incomplete inputs. Furthermore, considering that spectrum RSS typically exhibits rapid yet low-amplitude fluctuations, instead of relying on regression networks, we introduce a generative-model-based approach to produce more accurate predictions with preserving subtle RSS variations.

To effectively capture TSS dependencies from incomplete historical inputs, we first introduce a *TSS Attention-Based Condition Construction* (TSS-CC) stage comprising three parallel self-attention feature extractors, each dedicated to the TSS dimensions. The resulting features are then integrated by a cross-attention-based module to obtain a unified representation, which serves as conditional guidance for future spectrum prediction. Then, to predict future spectrum RSS data with fine-grained variations, we further introduce a *Latent Conditional Diffusion-Based Spectrum Prediction* (LCD-SP) stage which is conditioned on the representation obtained from the previous stage. The future spectrum RSS data generation is conducted in a latent space. By progressively corrupting latent variables through a forward process and subsequently denoising them via a learned reverse backward process, this stage enables detail-preserving prediction of future spectrum RSS data with fine-grained variations. The main contributions of this paper are summarized as follows.

- In this work, we propose a spectrum prediction method, particularly considering scenarios where historical spectrum observations are incomplete. Unlike most existing works that capture one or two types of dependencies from historical observations, we jointly capture TSS dependencies from incomplete historical inputs to provide more informative support for spectrum prediction. Furthermore,

instead of adopting regression-based architectures, we introduce a generative-model-based paradigm to preserve subtle variations in future RSS. To this end, we propose TSS-LCD, a two-stage network for accurate and detail-preserving future spectrum prediction under incomplete observations.

- Considering that extracting informative TSS dependencies becomes challenging under incomplete historical observations, we design a TSS-CC stage that employs separate self-attention branches to capture TSS features from partially observed inputs and then fuses them into a unified conditional representation for guiding future spectrum generation. Then, to achieve accurate prediction with fine-grained RSS variations preserved, introduce an LCD-SP stage that performs denoising-based generation in a compact latent space via LCD, thereby improving prediction fidelity and preserving fine-grained spectrum variations.
- We evaluate TSS-LCD on the AERPAW wideband spectrum dataset, using three fixed nodes (CC1, CC2, LW1) on the NCSU campus. Each node records RSS every 1 min for one week (10,080 time points) over 85-335 MHz, uniformly discretized into 250 frequency bins. Extensive experiments demonstrate that our method consistently achieves superior prediction accuracy and fine-grained detail preservation, confirming its effectiveness and robustness under various incomplete observation scenarios.

II. RELATED WORK

Driven by the increasing demand for intelligent spectrum access and proactive interference management, spectrum prediction has become a central research topic.

A. Traditional Model-Based Methods

Traditional model-based approaches, such as autoregressive integrated moving average (ARIMA) models, hidden Markov models (HMMs), and compressive sensing techniques, have demonstrated promising performance under specific assumptions and constraints [20], [21], [22], [23], [24], [25]. For example, in [20], Mosavat-Jahromi et al. adopted an ARMA model combined with a non-homogeneous Markov chain to predict spectrum occupancy, achieving low-complexity performance under stationarity conditions. In [21], Safari et al. developed a multi-step forecasting framework by integrating chaotic time series analysis with multi-scale singular spectrum analysis, thereby enhancing prediction accuracy. To address data incompleteness, Ding et al. proposed the Robust Online Spectrum Prediction (ROSP) framework in [22], which simultaneously performs matrix completion and temporal forecasting, enabling robust real-time prediction under missing or corrupted observations. In another line of work, Luo et al. [25] introduced a homotopy-enhanced HMM framework to predict future spectrum RSS data along UAV trajectories, thereby supporting temporal-spatial spectrum modeling. Despite simplicity and interpretability, these methods rely heavily on statistical assumptions, which may break down considering

highly dynamic and non-linear TSS characteristics within spectrum RSS data. Consequently, these methods struggle to model complex spectrum usage patterns, limiting their effectiveness in practical scenarios.

B. Deep Learning-Based Prediction Methods

To address this limitation, deep learning (DL) techniques have emerged as powerful alternatives, offering strong capabilities in modeling non-linear and multi-dimensional dependencies directly from historical spectrum observations.

1) *Temporal Dependency Modeling*: A line of work focuses on capturing temporal dependencies in spectrum evolution using sequential models such as RNNs, LSTMs, and Transformers [26], [27], [28], [29], [30]. In [26], Pan et al. designed a 3D Swin Transformer framework to extract local and global temporal features from spectrogram sequences. In [27], Li et al. combined singular spectrum analysis with an attention-enhanced Bi-LSTM to capture multi-scale temporal structures. In [28], Cui et al. integrated trend-seasonal decomposition and SSA with an attention-BiLSTM to handle non-stationary spectral dynamics. In [29], Geng et al. proposed a dual-stage BiLSTM with gated attention to refine temporal feature learning. In [30], Mootoo et al. developed EMForecaster, a TCN-based framework with distribution-free uncertainty modeling for fine-grained temporal prediction.

2) *Temporal-Spectral/Spatial Dependency Modeling*: In addition, some studies further incorporate frequency or spatial dependencies to enhance multi-dimensional representation learning [34], [35], [36], [37], [40], [44]. In [34], Li et al. presented TF2AN, an attention-based temporal-frequency fusion network aimed at predicting fine-grained spectrum energy levels by modeling intra-band temporal dynamics and inter-band frequency correlations. In [44], Guo et al. proposed DSIL, a deep spectrum incremental learning framework that integrates dynamic change detection with adaptive prediction to address spectrum concept drift. In [35], Li et al. introduced CL-MFGCN, a contrastive learning-based GCN framework with multiscale feature fusion, capturing both explicit and implicit temporal-frequency correlations in multichannel spectrum data. In [36], Zhang et al. proposed A-GCRNN, an attention-enhanced graph convolutional recurrent network that jointly models frequency-band interactions and temporal dependencies for multi-band prediction. In [37], Zhang et al. developed SAMS-GNN, a self-adaptive multi-scale GNN that dynamically captures evolving inter-band relationships and multi-scale temporal dependencies. However, most of these approaches focus on either temporal-frequency or temporal-spatial dependencies, without fully modeling the joint TSS dependencies embedded in historical spectrum RSS data.

3) *TSS Dependency Modeling*: In addition, several recent works have attempted to explicitly model TSS dependencies in a unified manner, achieving improved prediction performance by jointly extracting multi-dimensional TSS features from complete historical observations [41], [42], [43]. In [41], Pan et al. developed Dconv-Former, a hybrid CNNCTransformer architecture that integrates depth-wise convolution and self-attention to jointly capture TSS dependencies for fine-grained

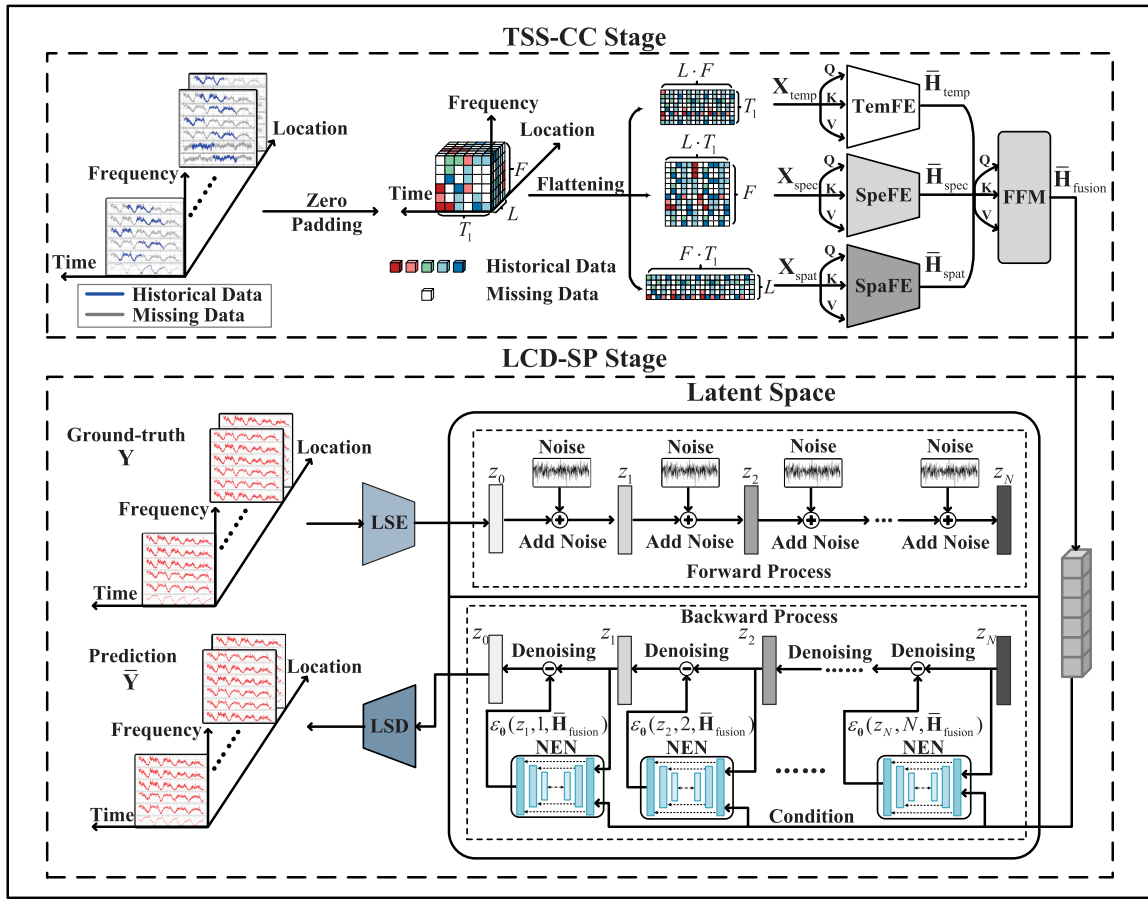


Fig. 3. Overview of the proposed TSS-LCD network.

are first zero-padded to construct a unified 3D tensor, where three axes correspond to the time slot, frequency bin, and location dimensions, respectively. Then, the tensor will be reshaped along each axis and fed into three parallel self-attention modules: Temporal Feature Extraction (TemFE), Spectral Feature Extraction (SpeFE), and Spatial Feature Extraction (SpaFE) to dig out implicit dependencies within each dimension, resulting in corresponding features. Finally, the extracted features are fused via a cross-attention based Feature Fusion Module (FFM), yielding a unified conditional representation which serves as informative guidance for further generation of future spectrum RSS data.

In the LCD-SP stage, the conditional representation guides a latent diffusion process to generate future spectrum RSS data. Specifically, the ground-truth future spectrum RSS data are first mapped into a latent space via a Latent Space Encoder (LSE), resulting in latent variables. Then, these latent variables will undergo a **forward process**, where Gaussian noise is progressively added over multiple diffusion steps to generate a sequence of noisy latent variables. These noisy latent variables are used to train a Noise Estimation Network (NEN), which takes the conditional representation and the noisy latent variables as inputs to estimate the added noise at each diffusion step. Once trained, the **backward process** under the guidance of the conditional representation starts from pure Gaussian noise and employs the NEN to iteratively denoise, recovering denoised latent variables of the future spectrum

RSS data. Finally, a Latent Space Decoder (LSD) is employed to reconstruct future spectrum RSS data from the denoised latent variables. Next, we will introduce the two stages in detail.

B. TSS Attention-Based Condition Construction

Since spectrum prediction relies on guidance of a condition containing underlying spectrum-usage patterns, it is critical to construct a conditional representation that learns these patterns by deeply mining the dependencies within historical spectrum observations. However, historical spectrum observations are inherently multi-dimensional and exhibit complex dependencies across the TSS dimensions. More importantly, missing values in historical observations may disrupt these inherent dependencies, making the construction of a reliable conditional representation substantially more challenging. To address this challenge, we design a TSS-CC stage, as illustrated in Fig. 4. Considering that spectrum RSS data exhibits informative and complex dependencies along the TSS dimensions, a single unified attention block may not sufficiently and consistently capture these dimension-specific patterns. Hence, to construct an information-rich conditional representation that better exploits such multi-dimensional dependencies, we adopt three parallel multi-head self-attention modules: TemFE, SpeFE, and SpaFE to separately capture long-range dependencies within the TSS dimensions, resulting in the corresponding features.

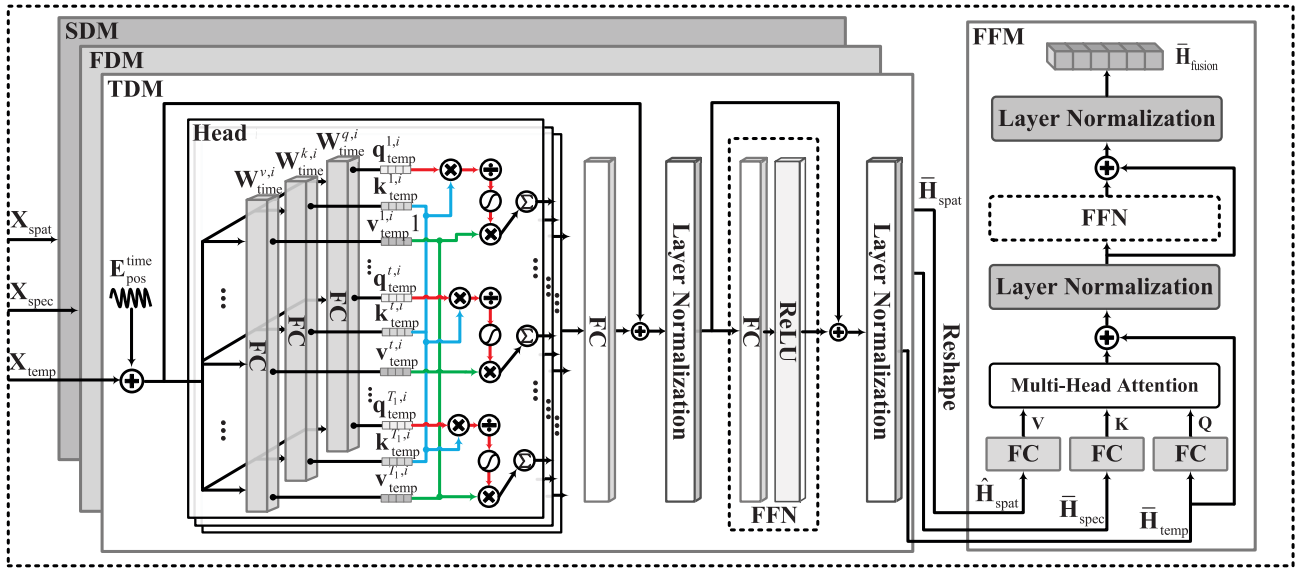


Fig. 4. TSS-CC stage.

By learning attention weights among observations along each dimension, these modules focus on valid and informative measurements while suppressing the impact of missing entries in incomplete historical observations. Then, to enable aggregation among these three dimensions and produce a unified condition representation, we introduce a cross-attention based FFM to integrate the extracted features. The details of each module are elaborated as follows.

1) *TemFE*: The TemFE module is used to capture the temporal dependencies within the incomplete historical observations, resulting in temporal features. Specifically, given the incomplete historical observations, we first perform zero-padding on the missing values to obtain a 3D input tensor $\mathbf{X} \in \mathbb{R}^{T_1 \times F \times L}$, where T_1 , F , and L denote the historical time slots, number of frequency bins, and number of locations, respectively. To enable temporal dependency modeling, we flatten the 3D tensor into a 2D matrix $\mathbf{X}_{\text{temp}} \in \mathbb{R}^{T_1 \times (L \cdot F)}$ denoted as

$$\mathbf{X}_{\text{temp}} = [\mathbf{x}_{\text{temp}}^1, \mathbf{x}_{\text{temp}}^2, \dots, \mathbf{x}_{\text{temp}}^{T_1}]^T \in \mathbb{R}^{T_1 \times (L \cdot F)}. \quad (1)$$

where $\mathbf{x}_{\text{temp}}^t \in \mathbb{R}^{1 \times (L \cdot F)}$, $t \in \{1, 2, \dots, T_1\}$ represents the incomplete historical observations over all locations and frequency bins at time slot t , which is also treated as a token in the TemFE module.

Since the multi-head self-attention mechanism processes spectrum data over all time slots in parallel, it lacks the ability to capture temporal order inherently. To address this limitation, we incorporate sinusoidal positional embedding into the input to compensate explicit temporal position information, described as

$$\tilde{\mathbf{X}}_{\text{temp}} = \mathbf{X}_{\text{temp}} + \mathbf{E}_{\text{pos}}^{\text{temp}}. \quad (2)$$

where $\mathbf{E}_{\text{pos}}^{\text{temp}} \in \mathbb{R}^{T_1 \times (L \cdot F)}$ is a sinusoidal positional embedding matrix.

The position-embedded input $\tilde{\mathbf{X}}_{\text{temp}}$ will be further fed into a multi-head self-attention block to compute the correlations between the spectrum data of all time slots. For each head

$i \in \{1, \dots, I\}$, the spectrum data $\tilde{\mathbf{x}}_{\text{temp}}^t \in \tilde{\mathbf{X}}_{\text{temp}}$, $t \in \{1, 2, \dots, T_1\}$ at each time slot t will be first projected into a query vector $\mathbf{q}_{\text{temp}}^{t,i}$, a key vector $\mathbf{k}_{\text{temp}}^{t,i}$, and a value vector $\mathbf{v}_{\text{temp}}^{t,i}$ through three independent fully connected (FC) layers, described as

$$\begin{aligned} \mathbf{q}_{\text{temp}}^{t,i} &= \tilde{\mathbf{x}}_{\text{temp}}^t \mathbf{W}_{\text{temp}}^{q,i}, \\ \mathbf{k}_{\text{temp}}^{t,i} &= \tilde{\mathbf{x}}_{\text{temp}}^t \mathbf{W}_{\text{temp}}^{k,i}, \\ \mathbf{v}_{\text{temp}}^{t,i} &= \tilde{\mathbf{x}}_{\text{temp}}^t \mathbf{W}_{\text{temp}}^{v,i}, \end{aligned} \quad (3)$$

where $\mathbf{W}_{\text{temp}}^{q,i}$, $\mathbf{W}_{\text{temp}}^{k,i}$, $\mathbf{W}_{\text{temp}}^{v,i} \in \mathbb{R}^{(L \cdot F) \times d_i}$ are learnable projection matrices, and d_i is the dimensionality per head.

Then, the attention score between time slots t and t' is computed by taking the scaled dot product between their corresponding query and key vectors. The score is then normalized with the softmax function to obtain attention weights as

$$a_{\text{temp}}^{t,t',i} = \text{softmax} \left(\frac{(\mathbf{q}_{\text{temp}}^{t,i}) (\mathbf{k}_{\text{temp}}^{t',i})^T}{\sqrt{d_i}} \right), \quad (4)$$

where $\text{softmax}(\cdot)$ denotes the standard normalization function.

Then, the output representation at each time slot t for head i is obtained by a weighted sum over the value vectors as

$$\mathbf{h}_{\text{temp}}^{t,i} = \sum_{t'=1}^{T_1} a_{\text{temp}}^{t,t',i} \cdot \mathbf{v}_{\text{temp}}^{t',i}, \quad (5)$$

where $\mathbf{h}_{\text{temp}}^{t,i} \in \mathbb{R}^{1 \times d_i}$. By stacking the outputs of all time slots, we obtain the representation for head i denoted as

$$\mathbf{H}_{\text{temp}}^i = [\mathbf{h}_{\text{temp}}^{1,i}, \mathbf{h}_{\text{temp}}^{2,i}, \dots, \mathbf{h}_{\text{temp}}^{T_1,i}]^T \in \mathbb{R}^{T_1 \times d_i}. \quad (6)$$

The outputs from all heads are then concatenated along the feature dimension and passed through a FC layer to produce the final multi-head attention output, described as

$$\mathbf{H}_{\text{temp}} = \text{Concat}(\mathbf{H}_{\text{temp}}^1, \dots, \mathbf{H}_{\text{temp}}^I) \mathbf{W}_{\text{temp}}^o, \quad (7)$$

where $\mathbf{W}_{\text{temp}}^o \in \mathbb{R}^{(I \cdot d_i) \times (L \cdot F)}$ is a learnable output projection matrix.

The multi-head attention output $\mathbf{H}_{\text{temp}} \in \mathbb{R}^{T_1 \times (L \cdot F)}$ is combined with the position-encoded input $\tilde{\mathbf{X}}_{\text{temp}}$ through a residual connection, followed by layer normalization:

$$\tilde{\mathbf{H}}_{\text{temp}} = \text{LayerNorm}(\tilde{\mathbf{X}}_{\text{temp}} + \mathbf{H}_{\text{temp}}). \quad (8)$$

Then, $\tilde{\mathbf{H}}_{\text{temp}}$ is passed through a position-wise feed-forward network (FFN), and another residual connection with layer normalization is applied:

$$\tilde{\mathbf{H}}_{\text{temp}} = \text{LayerNorm}(\tilde{\mathbf{H}}_{\text{temp}} + \text{FFN}(\tilde{\mathbf{H}}_{\text{temp}})), \quad (9)$$

where the FFN is defined as a two-layer fully connected network with ReLU activation. The final output $\tilde{\mathbf{H}}_{\text{temp}} \in \mathbb{R}^{T_1 \times (L \cdot F)}$ is the temporal feature.

2) *SpeFE and SpaFE*: The SpeFE module and the SpaFE module are used to capture spectral and spatial dependencies, respectively. Similar to the TemFE, these two modules perform multi-head self-attention along the spectral and spatial dimensions, resulting in spectral feature $\tilde{\mathbf{H}}_{\text{spec}} \in \mathbb{R}^{F \times (L \cdot T_1)}$ and spatial feature $\tilde{\mathbf{H}}_{\text{spat}} \in \mathbb{R}^{L \times (F \cdot T_1)}$, respectively.

3) *FFM*: Considering that the features extracted by the three separate modules are heterogeneous and not naturally aligned, directly using them as conditional guidance may not effectively exploit both the within-dimension dependencies and the cross-dimension information. Hence, to align and integrate these multi-dimensional features into a unified and informative conditioning signal, we further adopt a cross-attention-based FFM. Specifically, given the extracted features, the spatial feature $\tilde{\mathbf{H}}_{\text{spat}} \in \mathbb{R}^{L \times (T_1 \cdot F)}$ is first reshaped into $\hat{\mathbf{H}}_{\text{spat}} \in \mathbb{R}^{F \times (T_1 \cdot L)}$, ensuring that it aligns with the spectral feature $\tilde{\mathbf{H}}_{\text{spec}}$ in the number of rows, which is required for valid cross-attention computation. Then, the temporal feature, spectral feature, and the reshaped spatial feature are projected into the query, key, and value matrices via three independent FC layers, denoted as

$$\begin{aligned} \mathbf{Q}_{\text{fusion}} &= \tilde{\mathbf{H}}_{\text{temp}} \mathbf{W}_{\text{fusion}}^q, \\ \mathbf{K}_{\text{fusion}} &= \tilde{\mathbf{H}}_{\text{spec}} \mathbf{W}_{\text{fusion}}^k, \\ \mathbf{V}_{\text{fusion}} &= \hat{\mathbf{H}}_{\text{spat}} \mathbf{W}_{\text{fusion}}^v. \end{aligned} \quad (10)$$

where $\mathbf{W}_{\text{fusion}}^q \in \mathbb{R}^{(L \cdot F) \times d}$, $\mathbf{W}_{\text{fusion}}^k \in \mathbb{R}^{(T_1 \cdot L) \times d}$, $\mathbf{W}_{\text{fusion}}^v \in \mathbb{R}^{(T_1 \cdot L) \times d}$ are learnable projection matrices, and $d = L \cdot F$ denotes the cross-attention computation dimension.

Based on these projected matrices, the cross-attention output can be calculated as

$$\mathbf{H}_{\text{fusion}} = \text{softmax}\left(\frac{\mathbf{Q}_{\text{fusion}} \mathbf{K}_{\text{fusion}}^\top}{\sqrt{d}}\right) \mathbf{V}_{\text{fusion}}. \quad (11)$$

The cross-attention output is further processed by a residual connection and layer normalization, denoted as

$$\tilde{\mathbf{H}}_{\text{fusion}} = \text{LayerNorm}(\tilde{\mathbf{H}}_{\text{temp}} + \mathbf{H}_{\text{fusion}}), \quad (12)$$

followed by a feed-forward network (FFN) with another residual connection as

$$\tilde{\mathbf{H}}_{\text{fusion}} = \text{LayerNorm}(\tilde{\mathbf{H}}_{\text{fusion}} + \text{FFN}(\tilde{\mathbf{H}}_{\text{fusion}})), \quad (13)$$

The final output $\tilde{\mathbf{H}}_{\text{fusion}} \in \mathbb{R}^{T_1 \times (L \cdot F)}$ serves as the unified conditional representation for guiding the spectrum prediction in the next stage.

C. Latent Conditional Diffusion-Based Spectrum Prediction

In general, the spectrum observations exhibits dynamic and rapid but low-amplitude fluctuations, highlighting the fine-grained variations across the TSS domains. Hence, accurately spectrum prediction thus requires the ability to reconstruct these fine-grained variations. However, due to their small magnitude and rapid dynamics, these fine-grained variations are inherently difficult to reconstruct, as they are easily smoothed out or overlooked by models lacking sufficient detail-preserving capacity. To address this, we design the LCD-SP stage to recover such fine-grained variations guided by the conditional representation constructed in the previous stage. The LCD-SP stage consists of a forward process and a backward process. Next, we will introduce these two process.

1) *Forward Process*: In the forward process, a LSE module is first adopted here to project the ground-truth spectrum RSS data $\mathbf{Y} \in \mathbb{R}^{T_2 \times L \times F}$ into a latent variable $\mathbf{z}_0 \in \mathbb{R}^{1 \times d_{\text{latent}}}$, denoted as

$$\mathbf{z}_0 = \mathcal{E}(\mathbf{Y}), \quad (14)$$

where $\mathcal{E}(\cdot)$ denotes the transformation implemented by the LSE. Then, Gaussian noise is gradually added to the latent variable $\mathbf{z}_0$ over N diffusion steps, resulting in a sequence of noisy latent variables $\{\mathbf{z}_n\}_{n=1}^N$. At each step n , the noisy variable $\mathbf{z}_n$ is generated by

$$\mathbf{z}_n = \sqrt{\bar{\alpha}_n} \mathbf{z}_0 + \sqrt{1 - \bar{\alpha}_n} \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}), \quad (15)$$

where $\bar{\alpha}_n = \prod_{s=1}^n (1 - \beta_s)$ is the cumulative product of noise coefficients, and $\{\beta_n\}_{n=1}^N$ denotes a predefined variance schedule. Here, $\mathcal{N}(0, \mathbf{I})$ represents a standard multivariate Gaussian distribution with zero mean and identity covariance.

These noisy latent variables $\{\mathbf{z}_n\}_{n=1}^N$ are used to train a NEN to predict the Gaussian noise injected at each diffusion step. Specifically, the noisy latent variable $\mathbf{z}_n$ is injected into the NEN by *concatenation* with the conditional representation $\tilde{\mathbf{H}}_{\text{fusion}}$ and the diffusion step index n , denoted as

$$\bar{\boldsymbol{\epsilon}}_n = \boldsymbol{\epsilon}_\theta(\mathbf{z}_n, n, \tilde{\mathbf{H}}_{\text{fusion}}). \quad (16)$$

Here, θ denotes the learnable parameters of the NEN, which are optimized by minimizing the mean squared error between the estimated noise and true noise. The training details of the NEN will be further described in Subsection D.

2) *Backward Process*: In the backward process, the well-trained NEN is used to iteratively remove noise from a pure Gaussian noise $\mathbf{z}_N \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ to recover a denoised latent variable $\bar{\mathbf{z}}_0$. At each denoising step $n \in \{N, \dots, 1\}$, the denoising transition is modeled as a conditional Gaussian distribution:

$$p_\theta(\mathbf{z}_{n-1} | \mathbf{z}_n) = \mathcal{N}(\boldsymbol{\mu}_n, \sigma_n^2 \mathbf{I}). \quad (17)$$

Here, the mean of the conditional Gaussian distribution is computed as

$$\boldsymbol{\mu}_n = \frac{1}{\sqrt{\alpha_n}} \left(\mathbf{z}_n - \frac{1 - \alpha_n}{\sqrt{1 - \bar{\alpha}_n}} \bar{\boldsymbol{\epsilon}}_n \right), \quad (18)$$

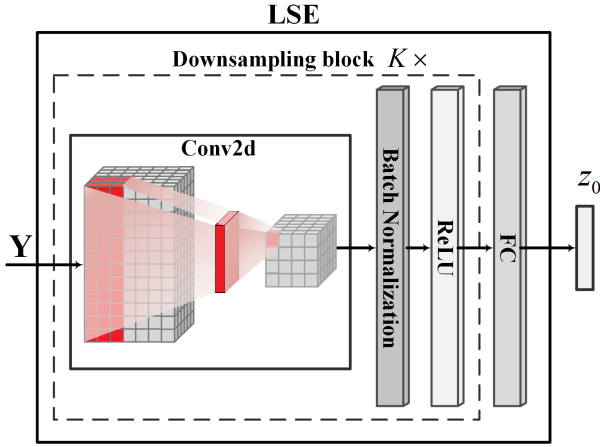


Fig. 5. Architecture of the LSE.

where $\alpha_n = 1 - \beta_n$. The variance of this distribution is given by

$$\sigma_n^2 = \frac{1 - \bar{\alpha}_{n-1}}{1 - \bar{\alpha}_n} \cdot \beta_n. \quad (19)$$

Accordingly, the noisy latent variable at step $n - 1$ can be generated by Eqn. (20), as shown at the bottom of the page.

By recursively applying Eqn. (20) from $n = N$ to 1, the denoised latent variable $\bar{z}_0$ can be obtained, which will be then passed through the LSD to obtain the predicted future spectrum RSS data, described as

$$\bar{Y} = \mathcal{D}(\bar{z}_0). \quad (21)$$

Next, we detail the architectures of the LSE, the NEN, and the LSD.

3) *LSE*: As illustrated in Fig. 5, the LSE module takes the ground-truth future spectrum RSS data $Y \in \mathbb{R}^{T_2 \times L \times F}$ as input. Then, the input will be processed by multiple stacked **downsampling blocks**, where each block contains a 2D convolutional layer (Conv2d), followed by Batch Normalization and a ReLU activation. These LSE blocks utilize Conv2d operations to progressively perform downsampling, effectively compressing the ground-truth future spectrum RSS data into a lower-dimensional representation. The output from the final block is then flattened and passed through FC layers to obtain the latent variable z_0 .

4) *NEN*: As illustrated in Fig. 6, the NEN takes the noisy latent variable z_n , the diffusion step index n , and the conditional representation $\bar{H}_{\text{fusion}}$ as inputs. These three components are first concatenated and reshaped into a 1D sequence, which is then processed by a 1D convolutional (Conv1D) based U-Net architecture composed of downsampling, bottleneck, and upsampling stages with skip connections. Finally, the NEN outputs the estimated noise $\bar{\epsilon}_n$ for step n .

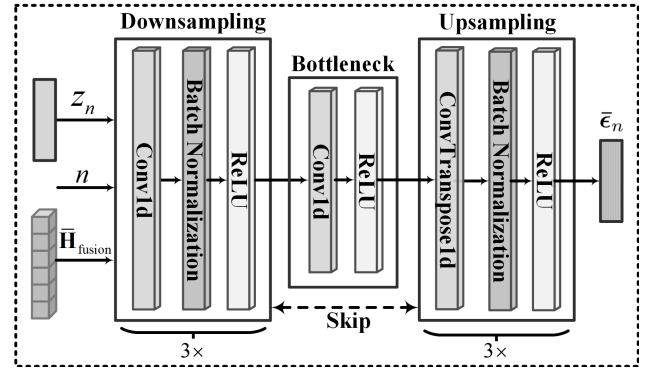


Fig. 6. Architecture of the NEN.

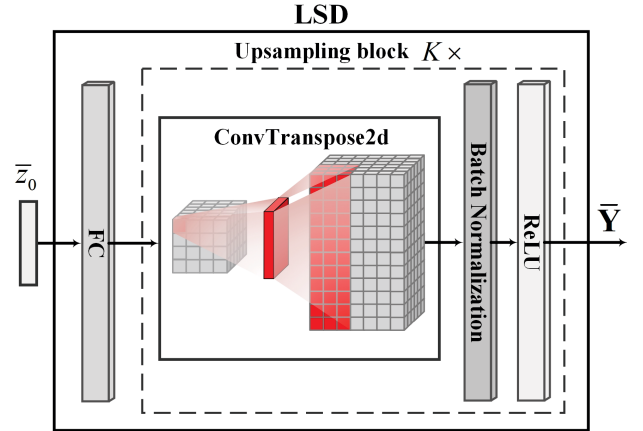


Fig. 7. Architecture of the LSD.

5) *LSD*: As illustrated in Fig. 7, the LSD is implemented as a stack of **unsampling blocks** that mirror the LSE structure in reverse. Specifically, the denoised latent variable $\bar{z}_0$ is first fed into a fully connected layer. Then, a sequence of unsampling blocks is applied, each consisting of a transposed 2D convolution (ConvTranspose2d), followed by batch normalization and ReLU activation. This hierarchical upsampling process gradually reconstructs and obtain the predicted spectrum RSS data $\bar{Y}$.

D. The Off-Line Training and On-Line Inference Procedure

The proposed TSS-LCD can be deployed in an **off-line training** and **on-line inference** workflow. In the off-line training phase, a large training dataset, in which each training sample contains the historical input X and its corresponding ground-truth Y , is used to train the TSS-LCD network. In the on-line inference phase, the well-trained network will take real-time spectrum observations as inputs to construct a conditional representation, which will then guide the backward

$$\begin{aligned} z_{n-1} &= \mu_n + \sigma_n \cdot \bar{\epsilon}_n \\ &= \frac{1}{\sqrt{\alpha_n}} \left[z_n - \frac{1 - \alpha_n}{\sqrt{1 - \bar{\alpha}_n}} \epsilon_{\theta}(z_n, n, \bar{H}_{\text{fusion}}) \right] + \sqrt{\frac{1 - \bar{\alpha}_{n-1}}{1 - \bar{\alpha}_n}} \cdot \beta_n \cdot \eta, \quad \eta \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \end{aligned} \quad (20)$$

process to progressively denoise to generate the prediction $\bar{\mathbf{Y}}$. Next, we will detail the two phases.

1) *Off-Line Training Phase*: The training of TSS-LCD is proceed in three stages: *Latent Autoencoder Pretraining*, *TSS Stage Training*, and *LCD Stage Training*.

a) *Latent autoencoder pretraining*: The LSE and LSD modules form a symmetric autoencoder that learns to compress and reconstruct complete future spectrum RSS data in a low-dimensional latent space. Given the ground-truth $\mathbf{Y}$, the autoencoder is trained by minimizing the reconstruction loss

$$\mathcal{L}_{\text{recon}} = \|\mathbf{Y} - \bar{\mathbf{Y}}\|_2^2 = \|\mathbf{Y} - \mathcal{D}(\mathcal{E}(\mathbf{Y}))\|_2^2, \quad (22)$$

where $\mathcal{E}(\cdot)$ and $\mathcal{D}(\cdot)$ denote the LSE and LSD, respectively.

b) *TSS stage training*: After pretraining the latent autoencoder, we separately train the TSS-guided conditional modeling part, i.e., TemFE, SpeFE, SpaFE, and FFM, to learn a robust conditional representation from incomplete historical observations. Specifically, given a training pair $\{\mathbf{X}, \mathbf{Y}\}$, the historical input $\mathbf{X}$ is processed by the three parallel feature extraction modules (TemFE, SpeFE, and SpaFE) and then fused by the FFM to obtain a unified conditional representation $\bar{\mathbf{H}}_{\text{fusion}}$. During this stage, the pretrained LSE/LSD are kept frozen, and the TSS modules are optimized to provide reliable conditional information for the subsequent generation stage.

c) *LCD stage training*: After the TSS modules converge, we train the LCD part, i.e., the NEN, in a separate stage. In this stage, the parameters of the pretrained LSE (and LSD) as well as the trained TSS modules are fixed. Given $\{\mathbf{X}, \mathbf{Y}\}$, we first compute $\bar{\mathbf{H}}_{\text{fusion}}$ from $\mathbf{X}$ using the fixed TSS modules, and encode the ground-truth $\mathbf{Y}$ into a latent variable $\mathbf{z}_0 = \mathcal{E}(\mathbf{Y})$ using the frozen LSE. Then, $\mathbf{z}_0$ undergoes the forward diffusion process to obtain the noisy latent variable $\mathbf{z}_n$ as in Eqn. (15). The conditional representation $\bar{\mathbf{H}}_{\text{fusion}}$, the diffusion step index n , and the noisy latent variable $\mathbf{z}_n$ are fed into NEN to estimate the noise $\bar{\epsilon}_n$, which is used to construct the denoising loss

$$\mathcal{L}_{\text{denoise}} = \|\epsilon - \bar{\epsilon}_n\|_2^2 = \|\epsilon - \epsilon_{\theta}(\mathbf{z}_n, n, \bar{\mathbf{H}}_{\text{fusion}})\|_2^2. \quad (23)$$

By minimizing $\mathcal{L}_{\text{denoise}}$, only the parameters of NEN are updated, enabling the diffusion model to leverage the fixed TSS conditional guidance to generate fine-grained predictions in the latent space.

2) *On-Line Inference Phase*: Once training is complete, the TSS-LCD network can be deployed for real-time prediction. Specifically, during the online inference phase, spectrum sensing nodes first perform partial-band scanning, resulting in a incomplete historical input $\mathbf{X} \in \mathbb{R}^{T_1 \times L \times F}$. This input is transmitted to a base station or edge server. Based on the well-trained TSS-LCD model deployed on the base station or the edge server, the unified conditional representation $\bar{\mathbf{H}}_{\text{fusion}}$ will be first constructed via the TSS-CC stage. Then, guided by the conditional representation, the backward process in the LCD-SP stage begins by sampling a pure Gaussian noise $\mathbf{z}_N \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. The model then denoises the latent variable $\mathbf{z}_N$ using the learned NEN as in Eqn. (20) from $n = N$ to 1. After N denoising steps, the recovered latent $\bar{\mathbf{z}}_0$ is passed through the LSD to reconstruct the prediction $\bar{\mathbf{Y}}$.

Algorithm 1 Training and Implementation of TSS-LCD

```

1: Offline Training Phase:
2: Stage I: Pretrain Latent Autoencoder
3: for each iteration do
4:   Encode future spectrum:  $\mathbf{z}_0 = \mathcal{E}(\mathbf{Y})$ 
5:   Reconstruct:  $\bar{\mathbf{Y}} = \mathcal{D}(\mathbf{z}_0)$ 
6:   Compute reconstruction loss:  $\mathcal{L}_{\text{recon}} = \|\mathbf{Y} - \bar{\mathbf{Y}}\|_2^2$ 
7:   Update parameters of LSE  $\mathcal{E}$  and LSD  $\mathcal{D}$ 
8: end for
9: Stage II: TSS Training.
10: Freeze  $\mathcal{E}$  and  $\mathcal{D}$ 
11: for each iteration do
12:   Extract condition:  $\bar{\mathbf{H}}_{\text{fusion}}$ 
13:   Update parameters of TemFE, SpeFE, SpaFE, and FFM
14: end for
15: Stage III: LCD Training.
16: Freeze TemFE, SpeFE, SpaFE, FFM, and  $\mathcal{E}$  (and  $\mathcal{D}$ )
17: for each iteration do
18:   Extract condition:  $\bar{\mathbf{H}}_{\text{fusion}}$ 
19:   Encode:  $\mathbf{z}_0 = \mathcal{E}(\mathbf{Y})$ 
20:   Sample diffusion step  $n \sim \{1, \dots, N\}$ 
21:   Sample noise:  $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
22:   Forward process:  $\mathbf{z}_n = \sqrt{\alpha_n} \mathbf{z}_0 + \sqrt{1 - \alpha_n} \epsilon$ 
23:   Predict noise:  $\bar{\epsilon} = \epsilon_{\theta}(\mathbf{z}_n, n, \bar{\mathbf{H}}_{\text{fusion}})$ 
24:   Compute loss:  $\mathcal{L}_{\text{denoise}} = \|\epsilon - \epsilon_{\theta}(\mathbf{z}_n, n, \bar{\mathbf{H}}_{\text{fusion}})\|_2^2$ 
25:   Update parameters of NEN
26: end for
27: Online Inference Phase:
28: Receive historical input:  $\mathbf{X}$ 
29: Extract condition:  $\bar{\mathbf{H}}_{\text{fusion}}$ 
30: Initialize:  $\mathbf{z}_N \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ 
31: for  $n = N, N-1, \dots, 1$  do
32:   Predict noise:  $\bar{\epsilon}_n = \epsilon_{\theta}(\mathbf{z}_n, n, \bar{\mathbf{H}}_{\text{fusion}})$ 
33:   Compute mean and variance:

$$\mu_n = \frac{1}{\sqrt{\alpha_n}} \left( \mathbf{z}_n - \frac{1 - \alpha_n}{\sqrt{1 - \alpha_n}} \bar{\epsilon}_n \right), \quad \sigma_n^2 = \frac{1 - \alpha_{n-1}}{1 - \alpha_n} \beta_n$$

34:   Denoising based on equation (20) to obtain  $\mathbf{z}_{n-1}$ 
35: end for
36: Decode:  $\bar{\mathbf{Y}} = \mathcal{D}(\bar{\mathbf{z}}_0)$ 

```

This offline training and online inference workflow allows real-time spectrum prediction, effectively leveraging incomplete historical observations to recover complete future spectrum RSS data with precise preservation of fine-grained variations. An overview of the entire training process is provided in Algorithm 1.

Remark 1: In practical dynamic wireless environments, the spectrum RSS distribution may evolve over time, leading to concept drift and potential degradation in both prediction accuracy and fine-grained detail preservation. Inspired by prior concept drift-aware studies showing that drift detection with online/periodic updating can mitigate the impact of non-stationarity [44], [45], TSS-LCD can be naturally complemented with drift-aware adaptation strategies without modifying the core TSS-CC and LCD-SP design.

V. SIMULATION RESULTS AND DISCUSSIONS

A. Settings

1) *Datasets and Implementation Details:* We evaluate the proposed TSS-LCD on a real-world wideband spectrum dataset collected from the AERPAAW platform, which contains RSS measurements at three fixed nodes, namely CC1, CC2, and LW1, deployed at three distinct locations within North Carolina State University [46].¹ These nodes are equipped with USRP B205mini devices running UHD1-based spectrum monitoring software. The measurements span one full week with continuous data collection, during which each node records the RSS every 1 minute.² In our experiments, we use a target frequency range from 85 MHz to 335 MHz, uniformly discretized into $F = 250$ frequency bins with a resolution of 1 MHz. The RSS values are measured in dBm.

The dataset contains $60 \times 24 \times 7 = 10,080$ time points per sensor. We adopt a sliding window strategy with sequence length $T_1 + T_2 = 60$, where the first $T_1 = 50$ time steps serve as historical observations and the following $T_2 = 10$ steps as the prediction ground-truth, resulting in 10,021 samples in total. A *training sample* is one historical observation and one ground-truth pair $(\mathbf{X}, \mathbf{Y})$ obtained by sliding a window of length $T_1 + T_2 = 60$ over the time series. The first $T_1 = 50$ minutes form the historical observations and the following $T_2 = 10$ minutes are the prediction ground-truth. These samples are divided into training and test sets using a 7:3 split ratio. Since the AERPAAW measurements are complete, we simply simulate incomplete historical observations by randomly selecting 25% of the historical inputs in each sample and setting them to zero.³ The future spectrum remains fully observed and is used as the prediction ground-truth.

The model is trained using the Adam optimizer with a learning rate of 1×10^{-4} , batch size of 32, and trained for 500 epochs on an NVIDIA RTX 3090 GPU. During training, early stopping is employed based on validation loss. All experiments are conducted under varying missing rates to evaluate robustness under incomplete observations. The architecture and hyperparameters of TSS-LCD are summarized in Table II.⁴

2) *Baselines and Evaluation Metrics:* To analyze the prediction results more directly, we use multi evaluation metrics

¹<https://aerpaw.org/dataset/february-2022-cc1-cc2-lw1-spectrum-measurements/>

²Although the default collection interval of the AERPAAW dataset is one-minute, the proposed TSS-LCD method can also support finer sampling intervals (e.g., second/millisecond-level) to meet different real-time communication decision-making requirements.

³Alternatively, incomplete observations can be simulated using a binary mask mechanism. Specifically, measurements in each sample are marked as observed (1) or missing (0). The masked input is fed into the model, so that the loss is computed only over the observed measurements while ignoring the contributions of missing ones, which helps the model avoid learning missing features.

⁴The diffusion step number is critical, as it governs the trade-off between prediction quality and computational cost. Specifically, using more diffusion steps generally produces higher-quality predictions that better preserve fine-grained RSS variations, but it also increases computation and sampling time; conversely, fewer steps reduce cost and latency but tend to lose details and degrade accuracy. Nevertheless, more steps are not always better, beyond a certain point, the gain in prediction quality becomes marginal while the computational cost continues to rise.

TABLE II
HYPER-PARAMETERS OF TSS-LCD

Parameter	Value
Historical length (T_1)	50
Prediction horizon (T_2)	10
Diffusion steps (N)	1000
Number of nodes (L)	3
Number of frequency bins (F)	250
Number of Transformer layers	2
Number of attention heads	4
Transformer hidden dimension (d)	1024
Dropout rate	0.1
Latent dimension	16
LSD and LSE hidden dimension	1024
Learning rate	1×10^{-4}
Batch size	32
Training epochs	500
Optimizer	Adam

to evaluate the performance of the proposed model, i.e., the Mean Squared Error (MSE), the Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Mean Absolute Percentage Error (MAPE), calculated as

$$\text{MSE} = \frac{1}{J} \sum_{j=1}^J (\hat{Y}_j - Y_j)^2, \quad (24)$$

$$\text{RMSE} = \sqrt{\frac{1}{J} \sum_{j=1}^J (\hat{Y}_j - Y_j)^2}, \quad (25)$$

$$\text{MAE} = \frac{1}{J} \sum_{j=1}^J |\hat{Y}_j - Y_j|, \quad (26)$$

$$\text{MAPE} = \frac{100\%}{J} \sum_{j=1}^J \left| \frac{\hat{Y}_j - Y_j}{Y_j + \delta} \right|. \quad (27)$$

where $\hat{Y}_j$ and Y_j denote the predicted and ground-truth values respectively, and J is the total number of prediction samples. A small constant δ is added in the MAPE denominator to avoid division by zero.

We compare the proposed method with the following three baseline models:

- ConvLSTM [39]: A hybrid model that combines CNNs and LSTMs for multi-channel, multi-step spectrum prediction at a single location.
- CSMA [41]: An Autoformer-based model that replaces the feed-forward network within the attention block with a channel-spatial attention module, enabling multi-channel, multi-step spectrum prediction across multiple locations.
- STS-PredNet [43]: An LSTM-based model tailored for spatiotemporal spectrum prediction, supporting multi-channel, multi-step forecasting across multiple locations.
- DSwinLSTM [40]: An encoder-decoder spectrum-map prediction framework built on SwinLSTM blocks, designed to jointly model local spatial dependencies and temporal dynamics.
- 3D-SwinSTB [26]: A lightweight Transformer-based model for spectrum prediction with 3D Swin-style spatiotemporal blocks, aiming at long-horizon forecasting.

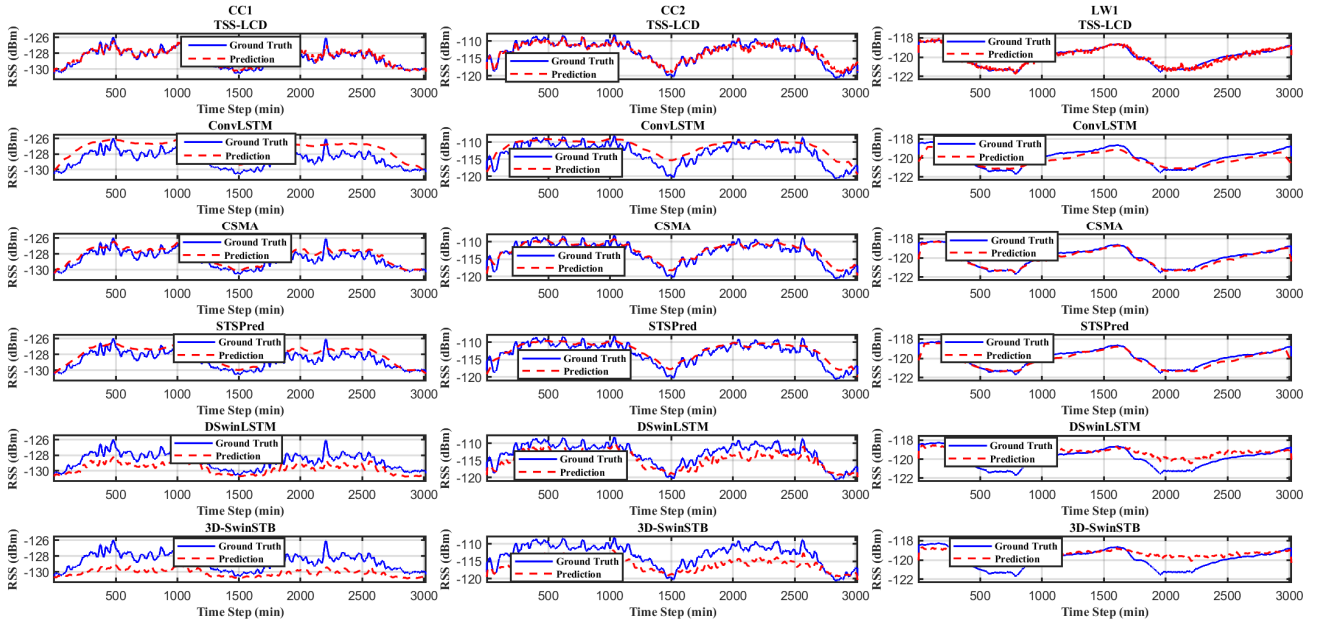


Fig. 8. Comparison of RSS curves across three locations (CC1, CC2, and LW1) using different algorithms.

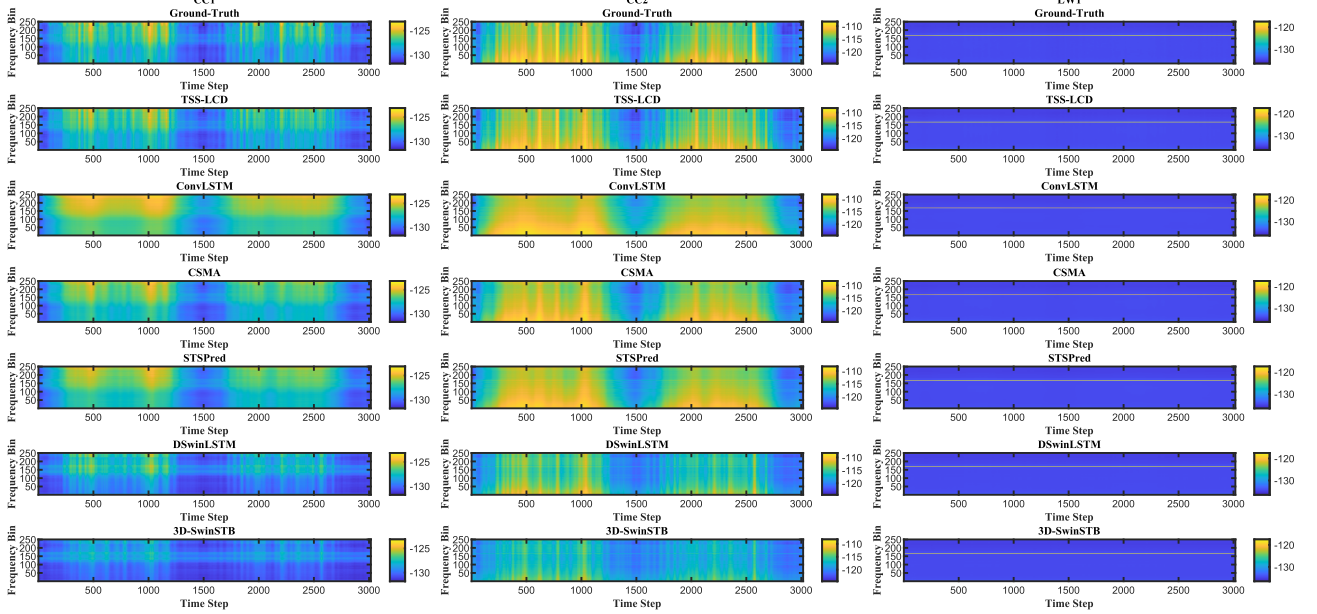


Fig. 9. Comparison of spectrum heatmaps obtained by four algorithms at three monitoring locations (CC1, CC2, and LW1).

B. Experimental Results and Discussions

To validate the effectiveness of our approach and its ability to make accurate predictions under incomplete observations, we conduct a comprehensive comparison of prediction curves, heatmaps, and error maps, as shown in Figs. 8–10. As shown in Fig. 8, we compare ground-truth and model-predicted RSS curves at the 85 MHz frequency bin for three sites (CC1, CC2, LW1). Our TSS-LCD attains the tightest overlap between the predicted and ground-truth RSS curves, preserving small-amplitude, short-term variations. By contrast, CSMA and STSPred generally track the coarse trend but show amplitude compression and timing mismatches around turning points, while DSwinLSTM, 3D-SwinSTB, and ConvLSTM exhibits the largest curve mismatch. While CSMA and STSPred do

attempt to capture TSS dependencies, they map extracted features directly to future RSS via one-shot regression, which tends to over-smooth and under-represent fine-grained fluctuations especially under incomplete observations. In contrast, DSwinLSTM, 3D-SwinSTB, and ConvLSTM mainly capture the temporal dependencies and cannot capture cross-frequency and cross-location dependencies, leading to the largest curve mismatch. In our framework, the TSS-CC stage jointly extracts TSS dependencies and forms a condition, which then guides an iterative denoising process in latent space (instead of one-shot regression). With this condition-guided iterative denoising, the model preserves small fluctuations, keeps peaks/valleys and turning points aligned, and achieves tight curve overlap under incomplete observations.

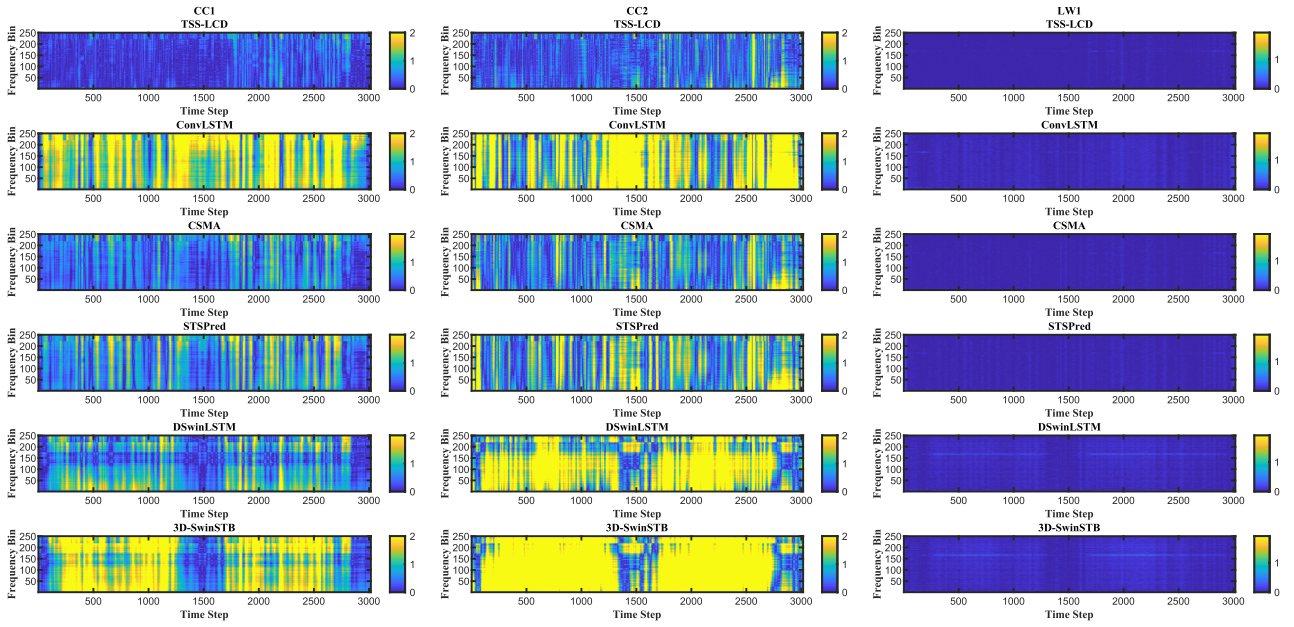


Fig. 10. Comparison of spectrum errormaps obtained by four algorithms at three monitoring locations (CC1, CC2, and LW1).

Then, we compare the time-frequency heatmaps with their absolute-error maps shown in Figs. 9 and Figs. 10, both reflecting the gap between ground truth and predictions. Across all sites, the heatmaps produced by TSS-LCD look essentially the same as the ground truth, whereas the baselines show noticeable differences, i.e., CSMA and STSPred, capture only the rough layout and appear over-smoothed, and DSwinLSTM, 3D-SwinSTB, and ConvLSTM deviates the most. The paired error maps confirm this: ours is the darkest with sparse errors, while the baselines contain large bright regions. Our method first jointly extracts TSS dependencies and forms a condition, then uses it to guide iterative denoising in latent space. This condition-guided process helps the model keep both the overall pattern and the details under incomplete observations, whereas CSMA/STSPred and time-only modeling (ConvLSTM) tend to smooth out details or miss structures.

In addition, the predicted curve may exhibit local volatility, with the prediction line appearing thicker and less stable than the comparative CSMA model. This is because that the diffusion-based generation in TSS-LCD starts from noise and iteratively denoises toward the final prediction, which is designed to preserve fine-grained spectrum variations. Intuitively, the stochastic sampling during denoising may leave minor noise that can appear as a slightly thicker prediction curve in relatively smooth scenarios (e.g., LW1). Nevertheless, the predicted curve aligns well with the overall trend of the ground truth, and such small fluctuations do not introduce noticeable bias or affect spectrum-state assessment (e.g., occupied/idle judgment) for proactive spectrum access decisions.

To further compare the effectiveness and accuracy of different methods, we conduct a quantitative evaluation based on four commonly used metrics: MSE, RMSE, MAE, and MAPE. As shown in Table III, our proposed method **TSS-LCD (Ours)** consistently outperforms all baselines across all metrics. Notably, it achieves an MSE and a MAPE, significantly lower than those of STS-PredNet, CSMA, DSwinLSTM,

TABLE III
COMPARISON OF PREDICTION ACCURACY ACROSS METHODS

Method	MSE↓	RMSE↓	MAE↓	MAPE↓
TSS-LCD (Ours)	0.1541	0.3926	0.2486	0.2075%
STS-PredNet [43]	0.9227	0.9605	0.5838	0.4917%
CSMA [41]	0.6441	0.8025	0.5029	0.4192%
ConvLSTM [39]	1.4065	1.1859	0.7333	0.6086%
DSwinLSTM [40]	1.4416	1.2007	0.8128	0.6867%
3D-SwinSTB [26]	3.9084	1.9770	1.3907	1.1754%

3D-SwinSTB, and ConvLSTM. This demonstrates the superior accuracy and robustness of our model in predicting fine-grained spectrum dynamics. The performance gap can be attributed to the modeling capability of different methods. DSwinLSTM, 3D-SwinSTB, and ConvLSTM mainly relies on recurrent operations to capture temporal dependencies and applies convolution to model frequency information. However, since it lacks a unified mechanism to jointly capture TSS dependencies, the method shows larger MSE/RMSE and higher MAE/MAPE, indicating greater prediction error. In addition, STS-PredNet and CSMA attempt to model joint TSS dependencies by applying convolutional or transformer-based feature extractors. While these methods improve over sequential models, they directly regress the prediction from extracted features, without leveraging any generative mechanism to refine or reconstruct the spectrum, which limits their ability to preserve high-frequency details and structural fidelity. In contrast, our TSS-LCD adopts a two-stage design: a TSS-CC stage to extract multi-dimensional dependencies, followed by a latent diffusion model that progressively generates the future spectrum in a compact latent space. This design not only ensures accurate modeling of TSS correlations but also enables high-fidelity spectrum reconstruction through iterative denoising, resulting in sharper, more realistic predictions.

TABLE IV
ABLATION STUDY ON KEY COMPONENTS OF TSS-LCD

Method	MSE↓	RMSE↓	MAE↓	MAPE↓
TSS-LCD (Ours)	0.1541	0.3926	0.2486	0.2075%
w/o TemFE	0.3027	0.5502	0.3515	0.2958%
w/o SpeFE	0.2910	0.5394	0.3512	0.2956%
w/o SpaFE	0.2861	0.5349	0.3371	0.2844%
w/o FFM	0.3139	0.5603	0.3615	0.3047%
w/o Latent Diffusion	0.4544	0.6741	0.4290	0.3638%
Diffusion in Original Space	0.3831	0.6189	0.3932	0.3325%

To investigate the effectiveness of each component in TSS-LCD, we conduct ablation experiments. As shown in Table IV, we remove, in turn, the TemFE, the SpeFE, the SpaFE, the FFM, and the latent diffusion module; we also compare latent-space diffusion with diffusion in the original data space. **Effect of TSS-CC Stage.** Removing any of the TemFE, SpeFE, or SpaFE modules leads to a significant performance degradation. Among them, discarding the temporal modeling (w/o TemFE) results in the most severe performance drop, increasing the MSE, which highlights the dominant role of temporal dependency in spectrum dynamics. The degradation caused by removing SpeFE and SpaFE also confirms the necessity of modeling frequency and spatial dependencies. As you can see, when removing the **SpaFE** module, the forecasting accuracy consistently degrades, demonstrating that SpaFE captures useful spatial relationships and provides informative support for future spectrum prediction under incomplete historical observations. Moreover, eliminating the FFM (w/o FFM), which integrates the three dependency representations via cross-attention, further worsens performance, indicating that independent modeling alone is insufficient and their joint integration is crucial for effective conditional representation.

Effect of LCD-SP Stage. We observe a notable increase in prediction errors when the latent diffusion module is removed, demonstrating the benefit of generating future spectrum RSS data in a compact latent space. Furthermore, replacing latent diffusion with a diffusion process directly conducted in the original spectrum space leads to suboptimal results, likely due to the increased complexity and noise sensitivity of high-dimensional prediction targets. These findings validate the advantage of latent-space modeling in preserving fine-grained patterns while mitigating generation difficulty. Overall, the ablation results confirm that both the TSS-CC and the LCD-SP jointly contribute to the superior performance of the proposed TSS-LCD framework.

Then, to evaluate the robustness of each model against incomplete historical observations, we conduct experiments with varying missing rates from 15% to 35% in the historical inputs. Specifically, we consider two missing patterns: (i) random missing, where a given proportion of historical observations is randomly selected and zero-padded; and (ii) continuous missing, where a contiguous time interval in the historical observations whose length is determined by the missing rate is randomly selected and set to zero. As shown in Fig. 11 and Fig. 12, all methods exhibit increased prediction errors as the missing rate rises, which is expected due to reduced temporal-frequency context. However, our

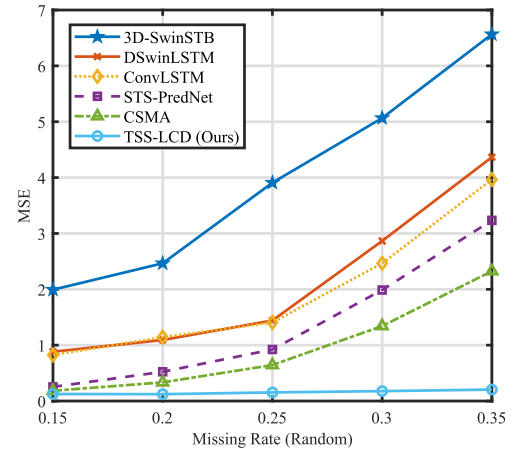


Fig. 11. Comparison of MSE performance under varying missing ratios for different models (Random Missing).

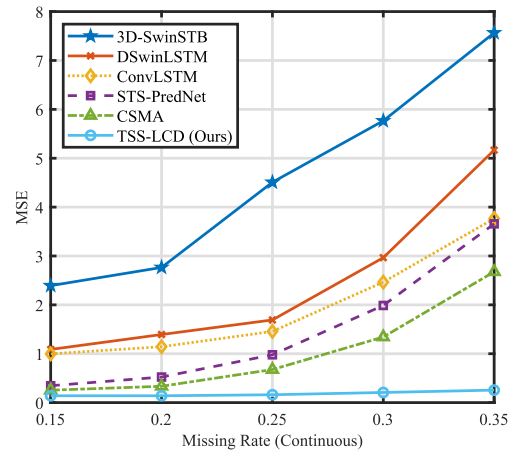


Fig. 12. Comparison of MSE performance under varying missing ratios for different models (Continuous Missing).

proposed method TSS-LCD consistently achieves the lowest MSE under all levels of data incompleteness, and its performance degrades much more slowly compared to STS-PredNet, CSMA, DSwinLSTM, 3D-SwinSTB, and ConvLSTM, in both missing patterns. This demonstrates the superior robustness of our model in handling missing observations. The TSS-guided attention mechanism enables effective utilization of partial information across time, frequency, and spatial dimensions, while the latent diffusion process enhances the model's ability to recover high-fidelity spectrum RSS data even under incomplete inputs with different missing patterns.

Finally, to further validate the effectiveness and robustness of each method in long-term prediction scenarios, we evaluate their performance under varying prediction horizons. Specifically, we increase the output length from 10 to 50 time steps and report the corresponding MSE results for each method. As illustrated in Fig. 12, the prediction error of all models increases with longer forecast horizons, which is consistent with the inherent difficulty of long-term spectrum prediction. However, our proposed method TSS-LCD demonstrates significantly slower error growth compared to the baselines. Even at a horizon of 50 steps, TSS-LCD maintains a relatively low MSE, while other methods such as STS-PredNet, DSwinLSTM, 3D-SwinSTB, and ConvLSTM show

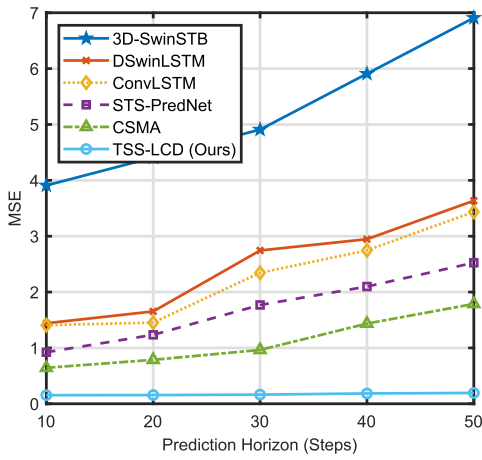


Fig. 13. Prediction performance under different forecast horizons. TSS-LCD maintains the lowest MSE and degrades more slowly with increasing prediction steps.

a sharp performance drop. This result confirms that our model not only performs well in short-term forecasting but also generalizes effectively to long-range prediction. The combination of TSS-aware encoding and latent diffusion modeling allows our method to capture long-term dependencies and generate stable, high-fidelity spectrum sequences across extended time intervals.

VI. CONCLUSION

In this paper, we presented TSS-LCD, a novel two-module framework for spectrum prediction with incomplete historical observations. By explicitly modeling TSS dependencies through a multi-branch attention mechanism, and by leveraging the generative capability of latent diffusion models, TSS-LCD is able to accurately reconstruct fine-grained future spectrum RSS data even under severely missing inputs. Experimental results on real-world datasets validate the superiority of TSS-LCD over existing baselines in terms of prediction accuracy, robustness to missing data, and long-range forecasting. Ablation studies further confirm the essential roles of both the TSS-attention module and latent diffusion generation module. This framework offers a promising solution for spectrum reconstruction under constrained sensing, and opens up new possibilities for proactive interference mitigation and adaptive spectrum management in dynamic wireless environments.

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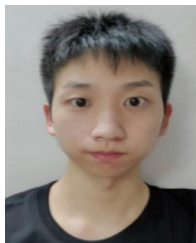


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